



# Library System

Borrowing, reservations, and circulation — built for small collections

Self-hosted · Open source

[github.com/uscabayaosj/LibrarySystem](https://github.com/uscabayaosj/LibrarySystem)

## Run a real lending library without buying library software.

Most library platforms are built for institutions with a systems librarian and a budget to match. This one is built for the shelf in the department office, the collection nobody has time to administer, and the borrower who only ever opens it on a phone.

**D**

### Department libraries

A few hundred to a few thousand titles, one or two staff, students borrowing between classes.

**P**

### Private collections

Personal or family libraries you actually lend from, and want returned.

**S**

### Small organizations

Churches, clinics, studios, NGOs, research groups — any shared collection with a sign-out sheet today.

## WHAT IT DOES

### Catalogue & copies

Title, author, ISBN, category, publisher, year, description. Tracks total copies against copies on the shelf, so availability is always accurate.

### Borrowing with real due dates

A configurable loan period with day-accurate tracking. Due today, due soon, and overdue are colour-coded everywhere they appear.

### Self-service renewals

Borrowers renew their own loans, subject to rules you set: not while overdue, not past the renewal limit, and never when someone else is waiting.

### Reservation queues

When every copy is out, borrowers join a queue and see exactly where they stand — "You're next in line" or "#3 in line". Filled automatically on return.

### Circulation desk

Librarian view for adding books, managing members, processing returns in one click, and reviewing full borrowing history with filters and pagination.

### Overdue visibility

Overdue items surface on both the borrower's dashboard and the librarian's, with configurable limits that block further borrowing.

## Designed phone-first, because that's how people borrow

Borrowers get a native-feeling mobile interface — a bottom tab bar, large titles that collapse as you scroll, and generated cover art for titles without a photo. It installs to the home screen as a real app. Librarians keep a full desktop workspace.

Add to Home Screen

Bottom tab navigation

Add due dates to phone calendar

Day-before reminders

30-day stay signed in

Light & dark

## REMINDERS WITHOUT A NOTIFICATION SERVICE

Every loan and reservation has an **Add to Calendar** button. One tap puts the due date into the borrower's own calendar app with an alarm the day before. No push infrastructure to run, no SMS bill, no email deliverability to manage — the phone does the reminding.



And books without a cover photo still look like books. Each title gets a distinct, stable colour tile generated from its ISBN — so shelves scan visually even when you have no cover images to upload.

MAKE IT YOURS — NO CODE, NO REDEPLOY

An administrator sets the name, logo, and theme colour from a settings page. The whole interface re-brands — sidebar, page titles, sign-in screen, and the installed app icon.

|                   |                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Organization name | Appears in the sidebar, browser titles, sign-in page, and the installed app name.                                                                                        |
| Logo              | Square PNG, JPEG, or WEBP — 512×512 to 1024×1024 px recommended, 2 MB max.<br>Regenerated automatically into the full app-icon set.                                      |
| Theme colour      | Pick one brand colour. Light and dark variants are derived automatically and checked against WCAG AA contrast, so an off-brand shade can't quietly make text unreadable. |

BUILT TO BE MAINTAINED BY SOMEONE WHO ISN'T A DEVELOPER

|                                                                                                                                                                                                        |                                                                                                                                                                       |                                                                                                                                                                                 |                                                                                                                                                                                                |
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| <b>Deploys from a link</b><br>Connect the repository to a host like Render or Vercel. A blueprint file (or a couple of environment variables) is the whole setup — no storage to provision separately. | <b>Upgrades apply themselves</b><br>Database migrations run automatically on every deploy. No manual database step, and no risk of code arriving ahead of its schema. | <b>Accessible by construction</b><br>Every colour pairing is verified against WCAG AA contrast in both light and dark modes, measured from rendered pixels rather than assumed. | <b>Works on poor networks</b><br>No external CDN, fonts, or icon services. The whole interface is served from your own instance, so it renders identically on a restricted or slow connection. |
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WHAT YOU NEED TO RUN IT

|             |                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------|
| Hosting     | Any container/VM host (Render, Railway, Fly.io) or serverless (Vercel). Free tiers are fine for evaluation. |
| Database    | Managed PostgreSQL (SQLite for local trials) — Render's own, or Neon.                                       |
| Storage     | None needed beyond the database. The uploaded logo lives there too.                                         |
| Maintenance | Push to deploy. Back up the database on your own schedule.                                                  |
| Licensing   | Self-hosted and open source. No per-seat fees, no vendor lock-in, and the data stays in your database.      |

IS IT THE RIGHT FIT?

|                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                               |
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| <b>✓ A good fit when</b> <ul style="list-style-type: none"><li>You're tracking a shared collection on paper, a spreadsheet, or trust</li><li>Borrowers are mostly on phones</li><li>One organization, one collection, one deployment</li><li>You want your own branding, not a vendor's</li><li>You'd rather own the data than rent access to it</li></ul> | <b>Look elsewhere if you need</b> <ul style="list-style-type: none"><li>MARC records, Z39.50, or union-catalogue federation</li><li>Barcode scanners and RFID hardware integration</li><li>Fines, payments, or accounting</li><li>Acquisitions, serials, or interlibrary loan</li><li>Multiple independent branches in one instance</li></ul> |
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GETTING STARTED

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| <b>1</b><br><b>Try it</b><br>Deploy to a free tier and load a few real titles — about an afternoon's work. | <b>2</b><br><b>Brand it</b><br>Set your organization's name, logo, and colour from the settings page. | <b>3</b><br><b>Invite borrowers</b><br>They register, sign in on a phone, and add it to the home screen. |
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SOURCE & DOCUMENTATION

github.com/uscabayaosj/LibrarySystem  
Includes a full deployment runbook, migration guide, and troubleshooting reference.

ENQUIRIES

Contact name

Email address · Phone